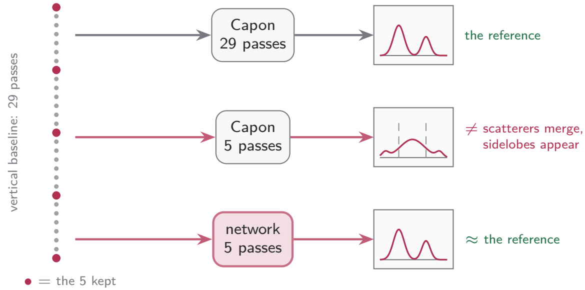
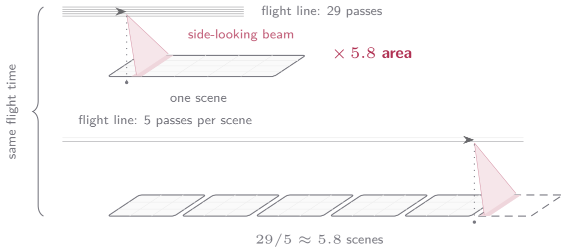
Motivation: more area from the same flight time

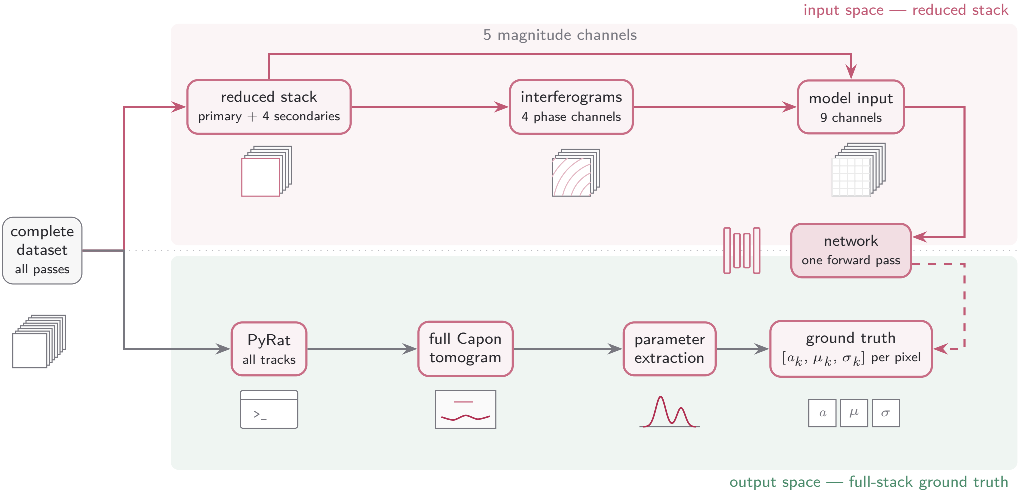




Elevation resolution is bought with flight passes: every pass adds one vertical baseline sample over the scene.

- **The problem**: on the reduced stack the classical inversion degrades: with 5 baselines nearby scatterers merge and sidelobes contaminate the profile.
- **The goal**: a network trained against full-stack ground truth recovers the full-stack parameters $[a_k, \mu_k, \sigma_k]$ from the 5-pass stack alone.
- **The payoff**: a scene then costs 5 passes instead of 29, so the same airtime surveys $29/5 \approx 5.8$ scenes.
- **The question**: how close does the reconstruction come to the reference? The rest of the talk measures that.

Methodology



A network learns to reconstruct, from the **reduced** stack alone,
the per-pixel profile parameters that the full-stack tomogram and classical fit produce.

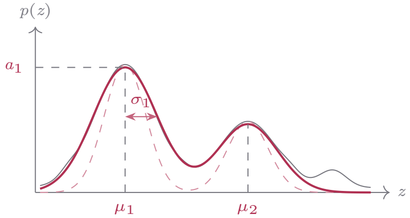
Ground truth: fitted Gaussian parameters

Supervised target: the per-pixel array $[a_k, \mu_k, \sigma_k]_{k=1}^{K}$ ($K{=}2$ today, $K{=}5$ in development), fit to the per-pixel max-normalised Capon spectrum $\tilde{p} = p/p_{\max}$ on the elevation grid ξ_n:

$$\mathcal{L} \;=\; \frac{1}{N}\sum_{n=1}^{N}\big(\hat{p}(\xi_n) - \tilde{p}(\xi_n)\big)^2, \qquad \hat{p}(\xi) = \sum_k a_k\, e^{-\frac{(\xi-\mu_k)^2}{2\sigma_k^2}}$$

One shared loss; each parameter group takes its own projected Adam step:

$$\begin{aligned} a_k &\leftarrow \big[\, a_k - \eta\,\mathrm{Adam}(\partial_{a_k}\mathcal{L})\,\big]_+ \\ \mu_k &\leftarrow \mathrm{clip}_{[\mu^-,\ \mu^+]}\big(\mu_k - \eta\,\mathrm{Adam}(\partial_{\mu_k}\mathcal{L})\big) \\ \sigma_k &\leftarrow \mathrm{clip}_{[\sigma^-,\ \sigma^+]}\big(\sigma_k - \eta\,\mathrm{Adam}(\partial_{\sigma_k}\mathcal{L})\big) \end{aligned}$$



Same pixel: Capon spectrum, peak initialisation (dashed), converged fit; the fitted $[a_k, \mu_k, \sigma_k]$ are the label.

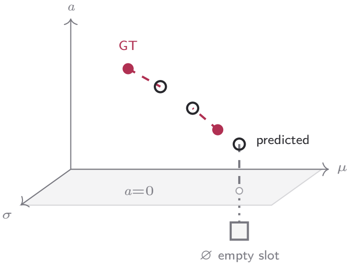
- Fitted in normalised space; the winning amplitudes are denormalised back, $a_k \to p_{\max}\, a_k$ (μ_k, σ_k already live on the physical axis).
- All pixels batched in one optimisation — not a per-pixel loop (next slide); most pixels activate one or two Gaussians (Act III).

Hungarian matching

- Each Gaussian is a **point in** (a, μ, σ) **space**: prediction and GT are two point sets; the matcher pairs them by distance and scores the *set*, not the slot layout.

$$\mathcal{L} = \min_{\pi \in S_K} \sum_{k=1}^{K} w_k \left\| \hat{\theta}_{\pi(k)} - \theta_k \right\|^2 \qquad \theta_k = (a_k, \mu_k, \sigma_k)$$

$w_k = \mathbf{1}[a_k > \tau]$ — GT *active mask*: 1 if present, 0 if empty.



Filled = active GT, open = predicted, $\varnothing$ = empty GT slot (below $a{=}0$). Its match (grey) is zero-weighted, so $\hat{\theta}_3$ is free.

GT θ_j

pred $\hat{\theta}_i$	θ_1	θ_2	$\varnothing$
$\hat{\theta}_1$	0.9	0.2	0.0
$\hat{\theta}_2$	0.4	0.8	0.0
$\hat{\theta}_3$	0.6	0.5	0.0

cost $C_{ij} = w_j \|\hat{\theta}_i - \theta_j\|^2$

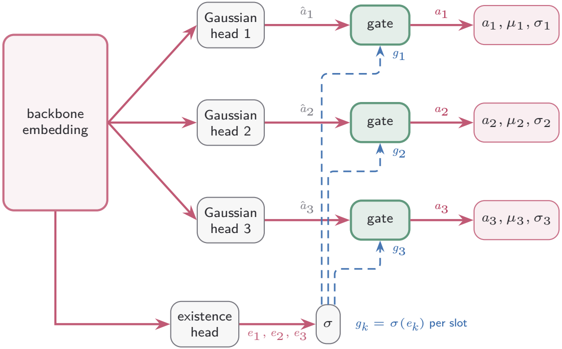
$\xrightarrow[\text{min cost}]{\pi^\star}$

0.9	**0.2**	0.0
0.4	0.8	0.0
0.6	0.5	**0.0**

$\sum C_{\pi^\star} = 0.6$

Column $\varnothing$ is a placeholder GT ($w_j{=}0$) — all-zero, so the leftover prediction matches it for free: $\hat{\theta}_1 \to \theta_2$, $\hat{\theta}_2 \to \theta_1$, $\hat{\theta}_3 \to \varnothing$.

Inside the set-prediction head — gate, blend, match

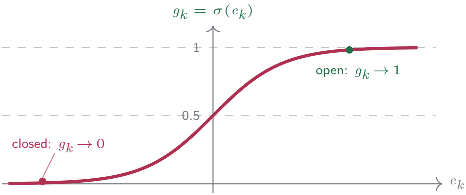


each gate: $a_k = g_k\,\hat{a}_k + (1 - g_k)\,a_{\mathrm{off},k}$

μ_k, σ_k pass through · a_{off} learned per slot

The gate is a sigmoid on the existence logit e_k:

- Open slots $(g_k \to 1)$: $a_k \approx \hat{a}_k$ — the raw regression survives.
- Closed slot $(g_k \to 0)$: $a_k \to a_{\mathrm{off},k} \approx 0$ *whatever* $\hat{a}_k$ is, collapsing onto the empty-set target at near-zero loss.
- e_k is internal — the head still emits $3K$ channels (a_k, μ_k, σ_k); a_{off} is a learned per-slot scalar (init 0).



Reading the reach — the network estimates statistics

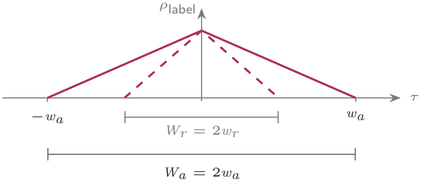
two labels at offset Δ split into shared and private looks:

$$w_a w_r \hat{\mathbf{R}}(x) = \sum_{\mathcal{W}\cap\mathcal{W}'} \mathbf{y}_p \mathbf{y}_p^{\mathsf{H}} + \underbrace{\sum_{\mathcal{W}\setminus\mathcal{W}'} \mathbf{y}_p \mathbf{y}_p^{\mathsf{H}}}_{\text{private to } x}$$

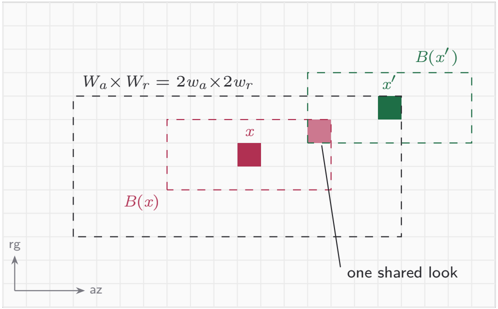
$$w_a w_r \hat{\mathbf{R}}(x') = \sum_{\mathcal{W}\cap\mathcal{W}'} \mathbf{y}_p \mathbf{y}_p^{\mathsf{H}} + \underbrace{\sum_{\mathcal{W}'\setminus\mathcal{W}} \mathbf{y}_p \mathbf{y}_p^{\mathsf{H}}}_{\text{private to } x'}$$

shared $\mathbf{S}(\Delta)$

the shared fraction at every offset τ is the correlation law:



the label correlates over $\pm w_a$ in azimuth and $\pm w_r$ in range — the centred patch spanning both triangles is $W_a \times W_r$.



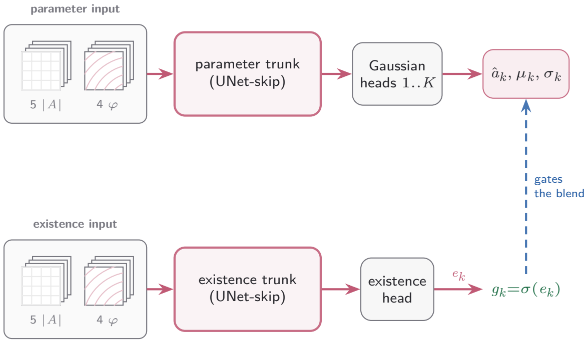
x' sits at the far corner of the patch: $B(x')$ keeps a single shared look with $B(x)$ — one step further and the windows are disjoint ($|\Delta_a| = w_a$ *or* $|\Delta_r| = w_r$).

The board at five seeds — the L1 family owns the top nine

#	model	loss	curve R^2 ↑	cosine ↑	SSIM elev ↑	SSIM range ↑	SSIM azim ↑	peak err ↓	recall ↑	2-sc recall ↑	score
1	unet-skip	L1-curve	**0.756** ±.012	0.924 ±.007	0.9676 ±.0004	0.9500 ±.0016	0.9611 ±.0008	1.29 ±.05	0.886 ±.006	0.728 ±.013	0.983
2	unet-skip	Charbonnier	0.751 ±.012	0.928 ±.004	0.9675 ±.0007	0.9499 ±.0024	0.9608 ±.0012	**1.25** ±.03	0.886 ±.004	0.727 ±.008	0.977
3	unet-skip	param-L1	0.741 ±.016	**0.931** ±.001	0.9669 ±.0004	0.9513 ±.0006	0.9608 ±.0005	1.38 ±.03	0.893 ±.004	0.745 ±.008	0.963
4	resunet	param-L1	0.734 ±.013	0.927 ±.007	0.9664 ±.0003	0.9502 ±.0006	0.9601 ±.0004	1.32 ±.07	0.894 ±.003	**0.746** ±.006	0.946
5	deeplab	param-L1	0.731 ±.006	0.923 ±.004	0.9663 ±.0003	0.9495 ±.0004	0.9597 ±.0003	1.35 ±.05	0.888 ±.002	0.736 ±.005	0.937
6	deeplab	L1-curve	0.731 ±.012	0.921 ±.013	0.9667 ±.0005	0.9491 ±.0016	0.9599 ±.0008	1.33 ±.07	0.873 ±.022	0.721 ±.007	0.935
7	resunet	L1-curve	0.737 ±.019	0.916 ±.007	0.9668 ±.0004	0.9488 ±.0007	0.9601 ±.0006	1.34 ±.06	0.881 ±.005	0.715 ±.007	0.930
8	resunet	Charbonnier	0.731 ±.016	0.921 ±.014	0.9668 ±.0006	0.9492 ±.0015	0.9601 ±.0006	1.31 ±.08	0.881 ±.006	0.716 ±.008	0.929
9	deeplab	Charbonnier	0.735 ±.008	0.913 ±.015	0.9668 ±.0005	0.9492 ±.0020	0.9602 ±.0008	1.62 ±.45	0.868 ±.037	0.722 ±.013	0.912
10	nafnet	Charbonnier	0.702 ±.009	0.906 ±.007	0.9645 ±.0004	0.9444 ±.0010	0.9569 ±.0004	1.48 ±.05	0.875 ±.008	0.711 ±.009	0.829
11	segformer	Charbonnier	0.688 ±.022	0.910 ±.016	0.9651 ±.0008	0.9447 ±.0028	0.9572 ±.0013	1.51 ±.20	0.866 ±.019	0.704 ±.019	0.829
12	nafnet	L1-curve	0.698 ±.007	0.908 ±.010	0.9646 ±.0005	0.9444 ±.0013	0.9568 ±.0007	1.43 ±.08	0.879 ±.005	0.714 ±.008	0.828
						⋮					
16	deeplab	MSE-curve	0.732 ±.010	0.872 ±.019	0.9615 ±.0005	0.9368 ±.0011	0.9527 ±.0004	1.62 ±.08	0.869 ±.004	0.710 ±.003	0.785
20	unet	L1-curve	0.660 ±.015	0.897 ±.026	0.9649 ±.0009	0.9441 ±.0043	0.9563 ±.0018	1.51 ±.28	0.870 ±.029	0.711 ±.020	0.760
21	unet-skip	param-MSE	0.647 ±.047	0.879 ±.017	0.9626 ±.0010	0.9398 ±.0032	0.9536 ±.0016	1.59 ±.13	0.881 ±.022	0.739 ±.014	0.692

runs: model · set-pred · Hungarian · $K{=}2$ · hv flips · presence A · loss · patch 64×32, five seeds; held-out band. Cells: mean ± seed std; **bold** = best mean in column (rows shown), marked only where best and worst differ by more than 5%. score = equal-weight mean of the five metric groups (curve error, variance, SSIM, cosine, peak), min–max scaled over the 35 arms; # = rank; ranks 13–15, 17–19, 22–35 on the next frame only. cosine = per-pixel profile cosine mean; peak err in elevation units; SSIM per axis vs the GT cube (denorm); recall = matched, 2-sc recall on GT $k{=}2$ pixels. green = the winner's score; red = the param-MSE reference cell, 21st of 35.

The dual model — two trunks, one head



each gate: $a_k = g_k\, \hat{a}_k + (1 - g_k)\, a_{\mathrm{off},k}$

the set-prediction head of Act III, unchanged — only the source of each embedding splits

- **Split the encoder, keep the head**: two independent UNet-skip trunks — the Gaussian heads read one embedding, the existence head the other. Gate blend and Hungarian matching stay as they are.
- **Each trunk picks its own stack**: the channel groups $|A|$ (5 amplitudes) and φ (4 interferograms) are selected per trunk — which modality reaches which head becomes an architectural choice.
- **Why route**: the input ablation split the labour — phase fires the slots, the full stack calibrates the parameters. The dual model can hand each head the input its job needs.

Pairing the objective — param-L1 plus a curve term

$$\mathcal{L}(w) = \left(\mathcal{L}_{\text{param-L1}} + w\,\mathcal{L}_{\text{curve}}\right) / (1 + w)$$

paired weight w

L1	0.01	0.05	0.25	0.5	1	magnitude
Charbonnier	0.01	0.05	0.25	0.5	1	
cosine	0.01	0.05	0.25	0.5	1	shape only

baseline — param-L1 alone ($w=0$)

1 baseline + 15 pair arms per K, five seeds each, at $K=2$ and $K=5$ = 160 runs

The ratio block left the $K=5$ gate under-firing at every split — more existence capacity did not wake it. This experiment changes the objective instead: keep param-L1 and add a synthesized-curve term.

- **Two signals, one loss**: param-L1 reaches a slot only through its matched pair; the curve term re-synthesizes the profile from *all* slots and is permutation-invariant — its gradient touches every slot, matched or not.
- **Magnitude or shape**: L1 and Charbonnier price the synthesized profile's magnitude; cosine normalizes both curves first and prices shape alone.
- **A normalized pair**: the two terms are averaged, so the validation loss changes meaning with w — arms compare on held-out metrics only, never on the loss.
- **The question**: if the conservative gate is starved of a profile-level error signal rather than capacity, a small w should move detection at $K=5$ and leave $K=2$ alone.

Next steps

- **Physical loss terms**: the five ranked terms on the next three frames: covariance matching, coherence re-synthesis, moments, total power, the Capon cycle.
- γ**-net adaptation**: adapting the γ-net architecture to this pipeline.
- **JEPA pretraining**: predicting in representation space, the closing frame.

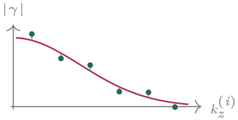
Data-consistency terms (the ranked top two)

Coherence re-synthesis

coherence domain

$$\gamma_P(k_z^{(i)}) = \frac{\sum_n P(\xi_n)\, e^{j k_z^{(i)} \xi_n}}{\sum_n P(\xi_n)}$$

$$\ell_{\text{coh}} = \left\langle \frac{1}{N_s} \sum_i \left| \gamma_P(k_z^{(i)}) - \gamma_T(k_z^{(i)}) \right|^2 \right\rangle$$



γ_P from the predicted mixture pulled onto the target coherences, baseline by baseline.

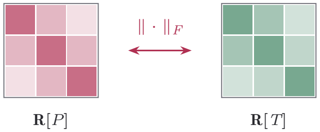
Cloude 2006 (PCT); same Fourier kernel as gamma-Net (Qian et al.).

Covariance matching

covariance domain

$$\mathbf{R}[P] = \mathbf{A}\, \mathrm{diag}(P)\, \mathbf{A}^H \Delta\xi$$

$$\ell_{\text{cov}} = \left\langle \frac{\|\mathbf{R}[P] - \mathbf{R}[T]\|_F^2}{\|\mathbf{R}[T]\|_F^2} \right\rangle$$



Synthesised vs target covariance — structure and relative scale, entry by entry.

COMET (Ottersten, Stoica, Roy 1998); TomoSAR: Cros & Ferro-Famil 2025; CS fidelity: Berenger et al. 2023.

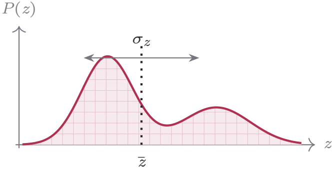
Descriptor terms: moments and total power

Moments

mass · phase-centre $\bar{z}$ · spread σ_z

$$m = \sum_n P(\xi_n)\,\Delta\xi$$

$$\bar{z} = \frac{\sum_n \xi_n P(\xi_n)}{\sum_n P(\xi_n)}, \qquad \sigma_z^2 = \frac{\sum_n (\xi_n - \bar{z})^2 P(\xi_n)}{\sum_n P(\xi_n)}$$



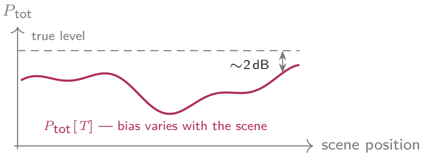
Centroid and spread are bias-robust *ratios* (Cros & Ferro-Famil 2025).

Total power

the one absolute-power quantity

$$P_{\text{tot}}[\,\cdot\,] = \sum_n (\,\cdot\,)(\xi_n)\,\Delta\xi$$

$$\ell_{\text{pow}} = \left\langle \left(P_{\text{tot}}[P] - P_{\text{tot}}[T]\right)^2 \right\rangle$$

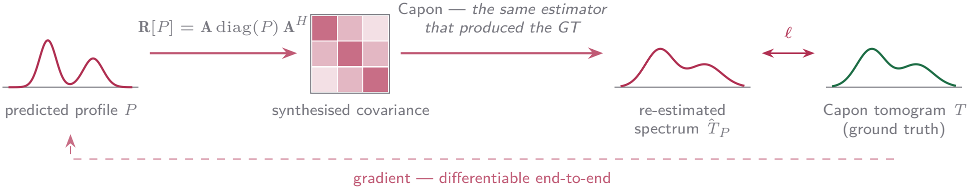


The target $P_{\text{tot}}[T]$ is not the true power: its bias moves with scene content (Lopez-Dekker & Mallorqui 2010) — the loss would imprint that bias.

The estimator-cycle term: Capon cycle

Push the prediction through the *same estimator* that produced the ground truth:

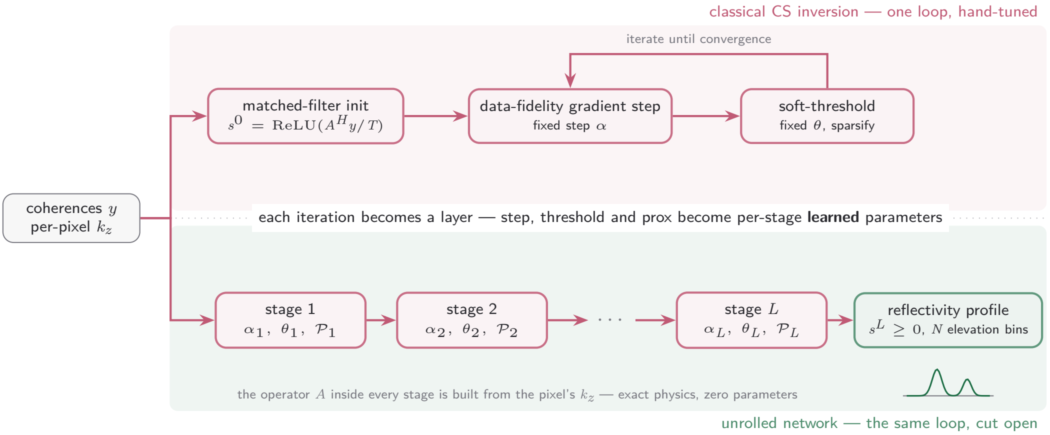
$$\hat{T}_P(\xi_n) = \left(\mathbf{a}^H(\xi_n)(\mathbf{R}[P] + \epsilon\bar{\sigma}\mathbf{I})^{-1}\mathbf{a}(\xi_n)\right)^{-1}$$



- Absorbs the Capon bias and the PSF **by construction**.
- Costs an $N_s \times N_s$ solve per pixel — the most expensive of the five.

Cycle-consistency lineage: Yaman 2020, Lu 2021, Vella & Mota 2020, Angermann 2023.

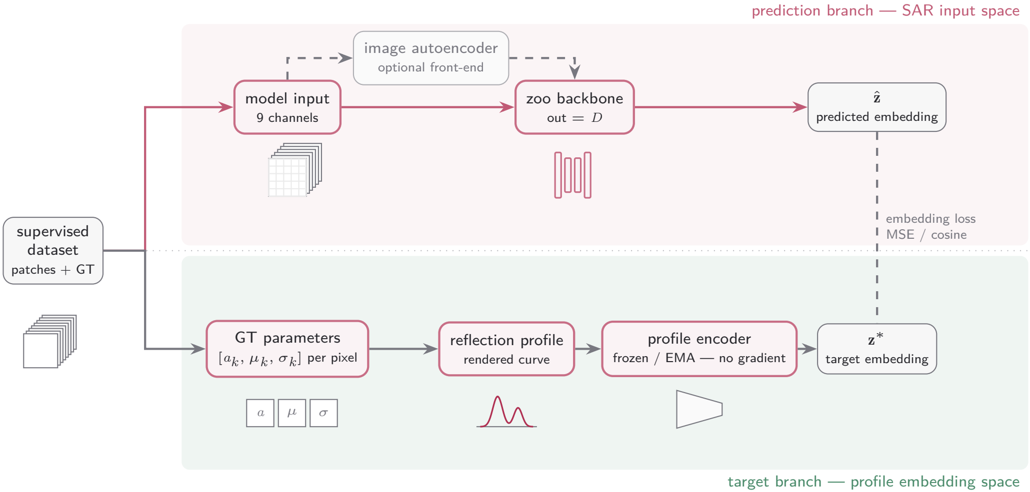
The unrolled model: the solver becomes the network



$$r^\ell = s^\ell + \alpha_\ell \, \mathrm{Re}\big[A^H(y - As^\ell)\big]/L_{\mathrm{lip}}, \qquad s^{\ell+1} = \mathrm{ReLU}\big(\mathcal{P}_\ell(r^\ell) - \theta_\ell\big)$$

γ-Net lineage (Qian et al., IEEE TGRS 2022); LISTA (Gregor & LeCun 2010); ISTA-Net (Zhang & Ghanem 2018). $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation; $L_{\mathrm{lip}} = TN\,\mathrm{d}z^2$ normalises the step. Default L=8: $\sim$1.4k parameters.

What JEPA does



A network learns to predict, from the reduced stack alone, the **embedding** that a separately-trained profile autoencoder assigns to the ground-truth reflection profile.